VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY INTERNATIONAL UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

**Data Science and Data Visualization**
**IT138IU**

# Proposal

**TOPIC: Interactive Nutrition & Personalized Meal Recommendation**

By Group TripleK – Member List

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# 1.Link GitHub

https://github.com/Nobodywinsbutme/CaloCola

# 2.Background and Motivation

As health issues like obesity and gout become more common, it is increasingly important to have accessible digital tools with friendly-visualization that help people track and understand their nutrition. To address this, this project centers on building an interactive Nutrition  and Meal Recommendation Dashboard designed to clearly map out the caloric density and nutrient breakdown of everyday meals.

By combining the Daily Food & Nutrition Dataset, VNDishes, and BMI/Obesity predictors, the dashboard gives users both a big-picture look at food properties and a tailored view of their own dietary needs.

The dashboard serves two main objectives:

- **Creating Personalized Nutrition Strategies:** The system calculates daily caloric needs based on a user's BMI, height, and weight. It then recommends meals tailored to specific goals—whether that is a high-protein diet for hitting the gym or low-calorie, high-volume meals for weight management.
- **Making Healthy Decisions Easier:** Using intuitive visualizations like physics-based Beeswarm plots, users can naturally explore the data. They can instantly compare the caloric density (Calories/gram) of different foods, like

meats versus vegetables, at a glance. This allows anyone to make smart dietary choices without needing a background in clinical nutrition.

# 3. Objectives of Data Visualization

**Exploring Caloric Density:** Instantly compare broad food groups (like meats versus fruits) to see which items pack the most calories per gram at a single glance.

**Category Drill-Downs:** Let users click into specific categories (like "Vegetables" or "Snacks") to filter the dashboard, allowing them to pinpoint exactly which foods are highest in calories, fats, or carbs.

**Personalized Progress Tracking:** Map a user's actual daily food intake directly against their required baseline, using their specific BMI and health profile.

**Smart Meal Recommendations:** Visually highlight the best food choices and meals that fit perfectly within a user's unique macro requirements and weight goals.

**Laying the Groundwork for AI (Future):** Design the interface with future features in mind—specifically, an upcoming tool where users can simply snap a photo of their plate, and the dashboard will automatically recognize the food and update their charts.

# 4.Data Collection

**Source Datasets:**

**Food Nutrition Dataset**

**thanh phan dinh duong.xls**

The project integrates multiple data sources from Kaggle and local repositories to create a comprehensive nutritional matrix:

1. **Daily Food & Nutrition:** The core dataset breaking down the exact macros (calories, fats, proteins, and carbs) for standard food categories.
2. **Calories in Food Items:** A massive list that helps standardize portion sizes across 100g, ounce, and per-serving measurements.
3. **VNDishes:** A specialized dataset mapping out the nutritional profiles of traditional Vietnamese meals, making the dashboard highly relevant to local users.
4. **BMI & Weight Change Datasets:** These provide the statistical baselines needed to power the dashboard's personalized health tracking and meal recommendations.

# 5.Data Preprocessing

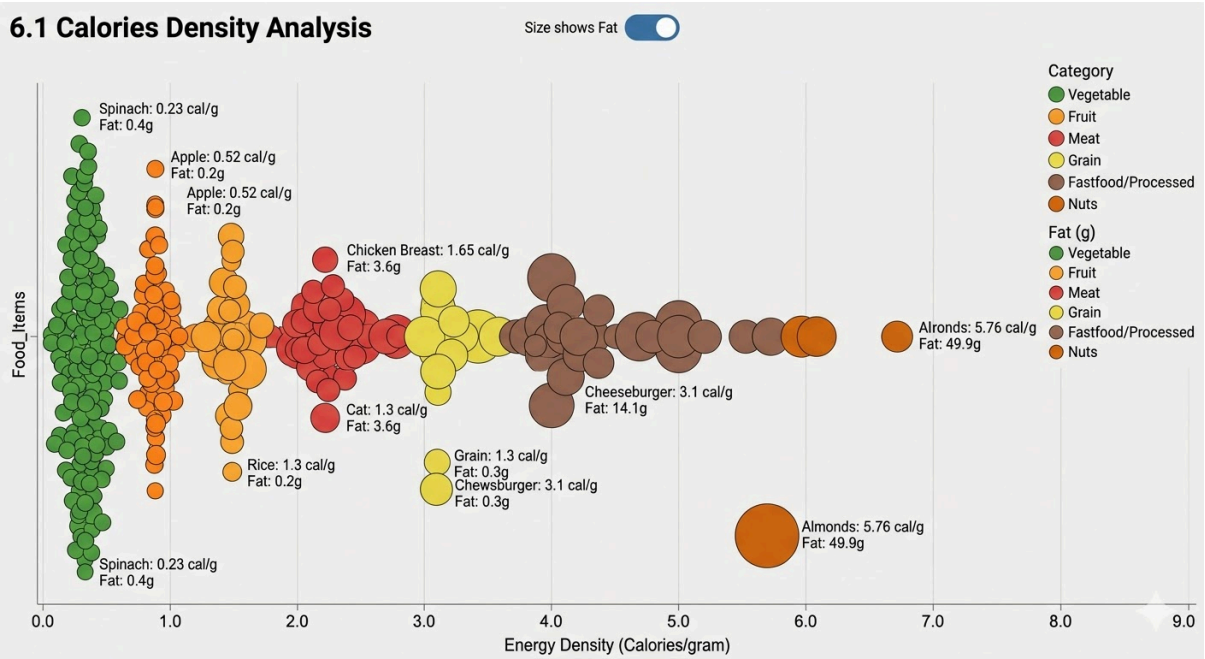
- **Standardization of Units:** Because the main focus of the dashboard is **"Caloric Density"** (Calories per gram), the first step is using Pandas to clean up the raw text. We will strip out labels like "58 cal" or "100g" and convert everything into clean, numeric float values so the density math is perfectly accurate.
- **Data Integration:** combining the global food datasets with the local VNDishes

data. This requires carefully aligning the columns (like Category, Calories, and Macros) so they merge seamlessly into a single, unified Pandas DataFrame.

- **Merging Data & Feature Reduction:** We will vertically concatenate the three food-related datasets into a 'Food Master Table'. The BMI dataset will be kept separate as a relational baseline. To optimize performance, we will drop duplicate food entries (prioritizing the local Vietnamese data), remove arbitrary index columns or unused micronutrient columns, and drop any records with unverifiable macronutrient profiles.
    - **What Will Be Merged: * The Food Master Table:** We will vertically concatenate (append) the *Daily Food & Nutrition*, *Calories in Food Items*, and *VNDishes* datasets. Before merging, we will align their core columns to a standard schema: Food_Name, Category, Serving_Size, Calories, Protein, Fat, and Carbs.
    - **Redundant & Duplicate Rows:** When merging global and local data, there may be overlaps (e.g., generic "Beef Noodle Soup" vs. authentic "Phở Bò"). We will drop the generic duplicates to prioritize the highly relevant *VNDishes* nutritional profiles.
- **Handling Missing Values:** Real-world datasets are rarely perfect, so we will filter out or fill in any missing macronutrient values. This guarantees that our Plotly charts—especially elements reliant on precise data, like bubble sizes—will render smoothly without throwing errors.

# 6. Visualization Design

## 6.1 Calories Density Analysis (Beeswarm Plot)



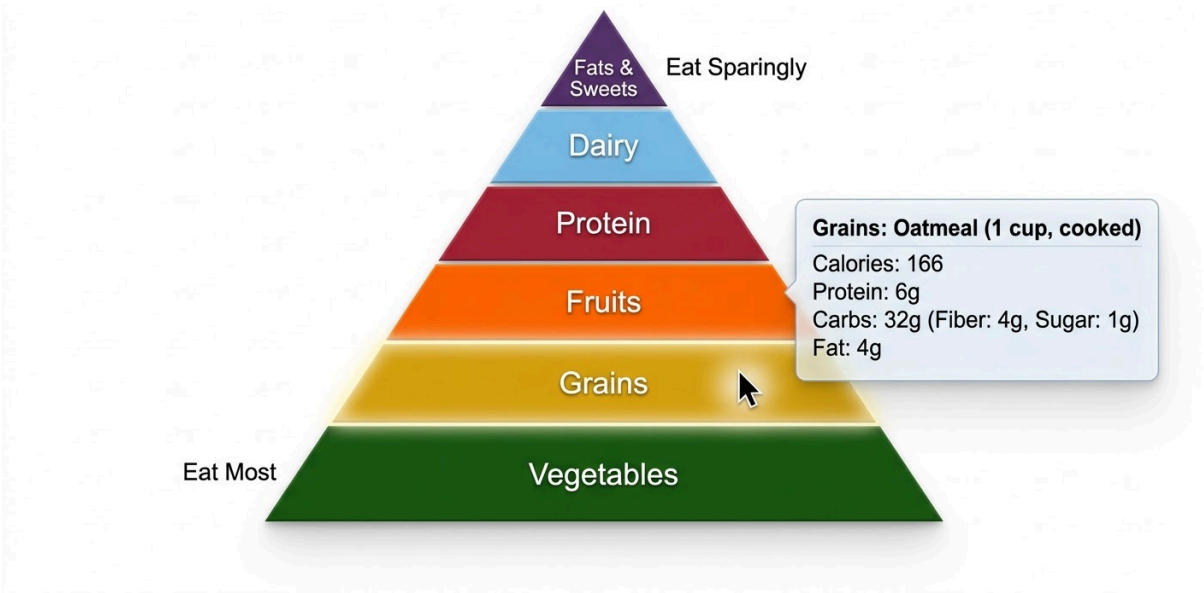

**Graph Variables**

The system will set up the following variables for the chart:

1. Node (circle/bubble): Represents a specific Food_Item in the database (e.g., 1 bubble is Potato, 1 bubble is Beef).
2. X-axis: Energy density (Calories/gram or Calories/100g). Low-calorie foods (Vegetables) will cluster on the left (e.g., 0.2 - 0.5 cal/gram) , while Nuts or Fastfood will drift towards the right (5 - 8 cal/gram).

3. Size (Radius): Represents the amount of Fat (g) or Carbohydrates (g). Users can use a Toggle button at the top to select "Size shows Fat" or "Size shows Carbs". Larger bubbles indicate foods high in fat or starch.
4. Color: Encoded by the Category field (Food classification).
    - Vegetable: Green
    - Meat: Red
    - Fruit: Blue or Orange
    - Grain: Yellow/Wheat
    - Fastfood/Processed: Brown or Purple
    - Nuts: Dark Orange

## 6.2 Recommended Food Pyramid (Pyramid Chart)

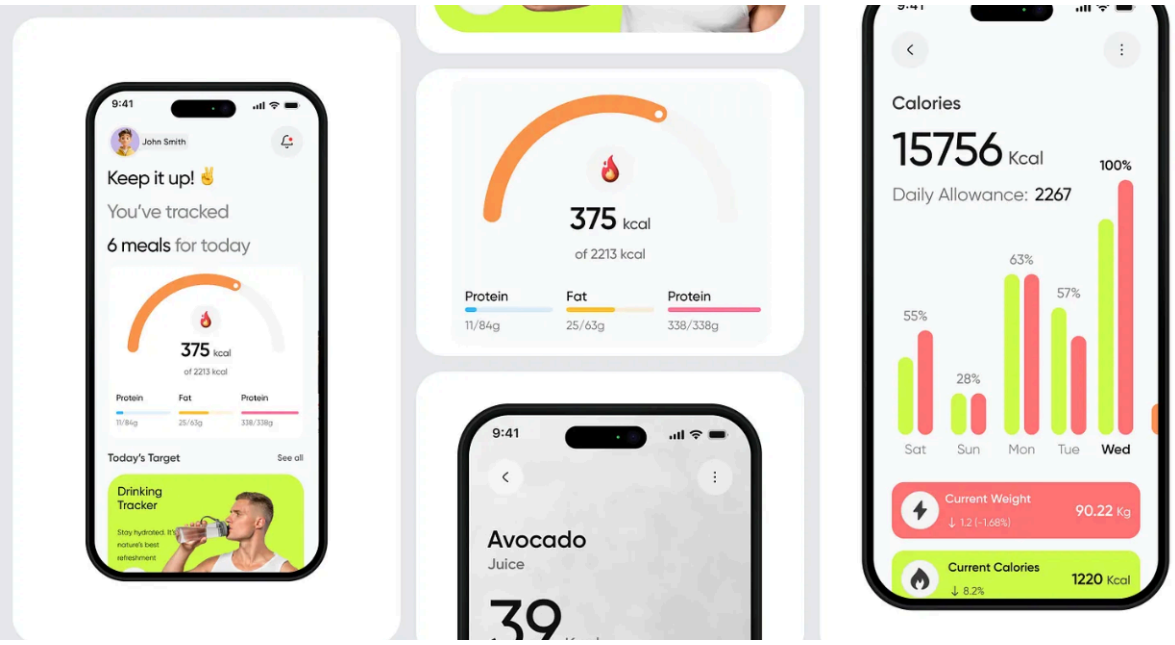



**Graph Variables**

The system will set up the following variables for the chart:

1. Layer/Tier (Horizontal slice): Represents different food categories based on dietary recommendations (e.g., eat most, eat more, eat moderately, eat less).
2. Position (Hierarchical Level): The vertical placement in the pyramid. The bottom layers contain foods recommended in large quantities (Vegetables, Grains), while the top layers contain foods that should be restricted (Sweets, High-fat foods).
3. Size (Area of the slice): Represents the recommended daily serving proportion for each specific food group.
4. Tooltip (Hover Interaction): Hovering over a specific layer or food item within the pyramid will display a popup with its exact nutritional components and total calories.

## 6.3 Daily Energy Tracking (Progress Rings, Linear Process and Weekly Checking)

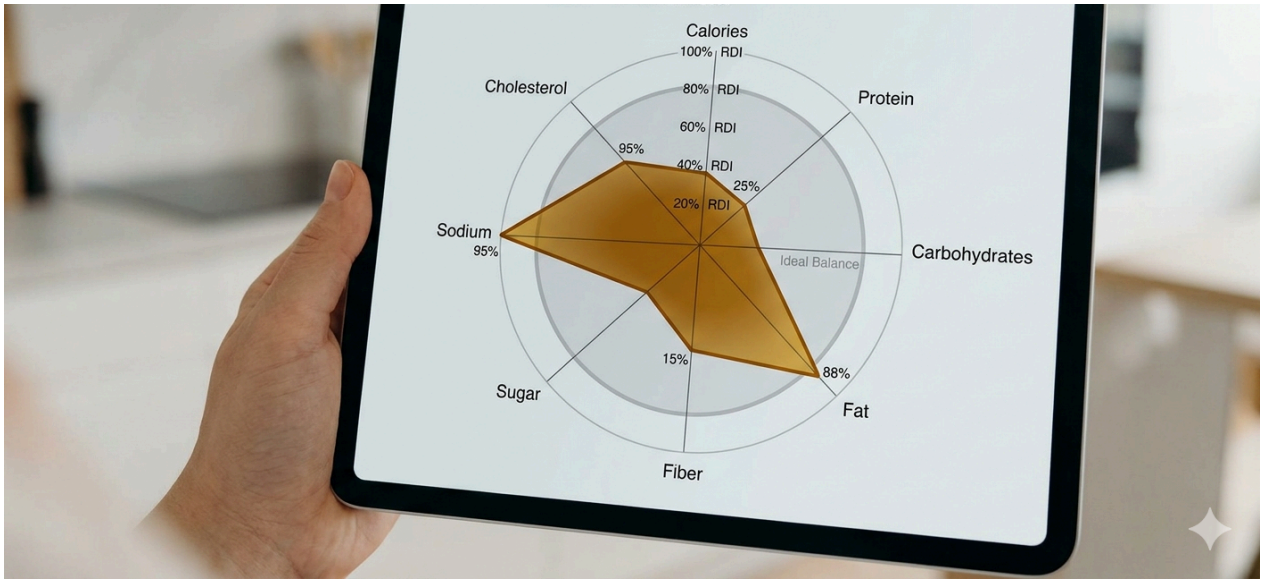


**Graph Variables**

The system will set up the following variables for the chart:

1. Process ring (Daily total calories): Represents the user's daily caloric intake progress compared to their personalized target (calculated via BMI, weight, and BMR formulas). The filled arc indicates the total calories consumed so far (e.g., 375 kcal), while the full half-circle represents the total daily allowance (e.g., 2213 kcal).
2. Linear Process Bar (Macronutrients):
    - Length: The filled portion of each separate horizontal bar indicates the actual amount of consumed macronutrients compared to their specific daily goal (e.g., 11/84g).
    - Color: Encoded by macronutrient type to easily visualize their progress:
        i. Protein: blue
        ii. Fat: yellow
        iii. Carbohydrates: Pink
3. Vertical Bar Chart (Weekly checking):
    - Bar: Represents the total daily calories consumed across the days of the week (Sat, Sun, Mon, Tue, Wed).
    - Color: The bars use conditional coloring (Green and Red) to likely indicate status (e.g., Green for within allowance, Red for exceeding allowance or specific threshold trends).

## 6.4 Meal Balance Assessment (Radar Chart)




**Graph Variables**

**The system will set up the following variables for the chart:**

1. Axes (Spokes): Represents the 8 core nutritional pillars tracked by the system (Calories, Protein, Carbohydrates, Fat, Fiber, Sugar, Sodium, Cholesterol).
2. Distance from Center: Indicates the quantity or percentage of the recommended daily intake for each specific nutrient.
3. Polygon (Shape): The connected data points form a shape that instantly visualizes the nutritional balance of a meal. A highly skewed shape (e.g., spiking high at Sodium and Fat, but dipping low at Fiber) serves as a visual warning of an unbalanced meal.

4. Color/Fill: A translucent color fill is applied to the polygon, allowing users to overlay multiple meals or compare their actual intake against an "ideal" balanced shape.

# 7.Must-have and Optional Features

## 7.1 Must-have Features

**The Main Beeswarm Chart:** This is the centerpiece of the dashboard. Instead of overlapping dots, it uses physics to naturally spread out thousands of food items so users can easily compare the caloric density of completely different groups, like meats versus vegetables, all on one screen.

**Personalized BMI & Calorie Calculator:** A simple, clean input panel where users punch in their basic stats (height, weight, age, gender). The app runs the math behind the scenes to find their Basal Metabolic Rate (BMR) and sets a custom daily calorie goal.

**Quick Category Filters:** Interactive dropdowns and toggles that let users slice the dataset instantly. They can filter the chart to show only traditional Vietnamese dishes or zero in on specific groups like "Beverages" or "High-Protein Foods."

**Detailed Hover Info:** Since we are packing a lot of data points onto the screen, hovering over any single bubble will pull up a clean tooltip showing exactly what that food is, its category, and its exact fat, carb, and protein stats per 100g.

## 7.2 Optional Features

These are the stretch goals. They aren't strictly necessary for the baseline grade, but they add a massive layer of polish and technical depth if you have extra time before the deadline:

**Automated Meal Construction:** An algorithmic feature that takes the user's calculated calorie limit and automatically highlights a combination of foods on the chart that would make a perfectly balanced daily meal plan.

**Animated Sorting Transitions:** Instead of the chart instantly snapping to a new layout when a user applies a filter, the bubbles will smoothly animate and float to their new X/Y axes, making the data exploration feel incredibly fluid.

**Future AI Integration Mockup:** A placeholder interface or proof-of-concept demonstrating where the future "Snap a Photo" AI food recognition feature will live, showing how it would seamlessly update the user's daily tracker.